Probability and Random Events

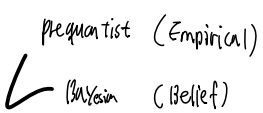
Jisang Yu

LESE152

Probability: Several Concepts

- Probability, Outcomes, and Events
- Rules of Probability
- Joint and Conditional Probability
- Independence
- Law of Total Probability and Bayes' Law
- Practice Questions

Probability



likelihood & Event
Random
→ deterministic

- Probability: A numerical measure of the likelihood that an event will occur. It ranges from 0 (impossible) to 1 (certain).
- Probability is a way of mathematically modeling a random process, so that we can understand it and make predictions about its future results.
- Casinos, banks, insurance companies, and any other businesses that manage risks rely on probability.
- Relationships between samples and populations are defined in terms of probability.

Assigning Probabilities

$P(A)$ = probability of A

$P(\text{dice} = 2) = \frac{1}{6}$

If) Rolling a dice: 1, 2, 3, 4, 5, 6

Sum of the numbers you get by rolling two different dice. $P(X=3) = \frac{2}{36}$

- Classical probability: If all outcomes are equally likely and the sample space is finite,

$$P(A) = \frac{\text{Number of outcomes favorable to } A}{\text{Total number of equally likely outcomes}}.$$

For example, $P(\text{Head}) = 1/2$ for a fair coin and $P(\text{Rolling a 5}) = 1/6$ for a fair die. What about $P(\text{Drawing an Ace from a deck of 52 cards})$? $4/52 = 1/13$.

- Empirical (relative frequency) probability: The observed frequency of A divided by the total number of trials. As the number of trials increases, the relative frequency stabilizes around the true probability (the Law of Large Numbers).
For example, if 47 out of 500 widgets are defective, $P(\text{Defective}) = 47/500 = 0.094$.

What Do Probabilities Mean?

- Frequentist interpretation: The random process can be repeated many times, and the probability of an event is the approximate fraction of times that it occurs. If you bet on Red in roulette 1000 times, you will win about 486 times.
- Bayesian interpretation: The random process is a one-time occurrence, and the probability represents the strength of our belief that the event will happen.
- The frequentist interpretation suits repeated settings like casino games or car insurance. The Bayesian interpretation makes more sense for predicting election results.

Roulette: An Example

- A roulette wheel has slots numbered 0 to 36. Slot 0 is green, 18 slots are red, and 18 slots are black.
- A ball is spun and lands in one slot. Players place bets such as:
  - Red (the ball lands in a red slot): pays \$1 per \$1 bet.
  - Black (the ball lands in a black slot): pays \$1 per \$1 bet.
  - A single number (the ball lands on that number): pays \$35 per \$1 bet.
- A roulette game is a random process. Something will happen, it matters to the players and the casino what will happen, but we do not know in advance what will happen.

Outcomes and the Sample Space

X = Number of 'Head's from 2 coin toss

Outcomes: 0, 1, 2

Simple space

- Outcome: A complete description of the result of a random process. Everything we are interested in should be defined in terms of the outcome.
- Sample space (Ω or S): The set of all possible outcomes. It must be exhaustive, and its outcomes must not overlap.
- Roulette: The outcome is the number of the slot in which the ball lands, b. The sample space is $\Omega = \{0, 1, 2, \ldots, 36\}$, which has $|\Omega| = 37$ elements.
- Other examples: $S = \{\text{Heads}, \text{Tails}\}$ for a coin flip; $S = \{\text{Defective}, \text{Non-Defective}\}$ for a product quality test; $S = \{2, 3, \ldots, 12\}$ for the total of two dice.

Events

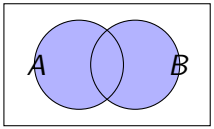
- Event (A): A subset of the sample space. Equivalently, a statement about the outcome that is either true or false.
- Simple (elementary) event: An event that contains only one outcome, e.g., rolling a 4 on a die, $\{4\}$.
- Compound event: A combination of two or more simple events, e.g., rolling an even number, $\{2, 4, 6\}$.
- Events in roulette:
  - The ball lands on 14: $b \in \{14\}$
  - The ball lands on red: $b \in Red = \{1, 3, 5, 7, 9, 12, 14, 16, 18, 19, 21, 23, 25, 27, 30, 32, 34, 36\}$
  - The ball lands on black: $b \in Black = \{2, 4, 6, 8, 10, 11, 13, 15, 17, 20, 22, 24, 26, 28, 29, 31, 33, 35\}$

Relationships Among Events

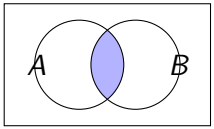
- Identical ($A = B$): The two events contain exactly the same outcomes. "The ball lands on 14" and "a bet on 14 wins" are identical.
- Implies ($A \subset B$): All outcomes of A are also in B. "The ball lands on 14" implies "the ball lands on red" since $\{14\} \subset Red$.
- Disjoint, or mutually exclusive ($A \cap B = \emptyset$): The two events share no outcomes, so they cannot both happen. $Red \cap Black = \emptyset$.
  But disjoint events can both fail to happen: if $b = 0$, neither red nor black wins.
- Any two elementary events are either identical or disjoint.

Venn Diagrams

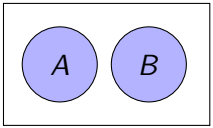
Union $A \cup B$



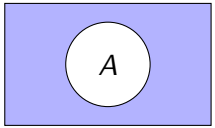
Intersection $A \cap B$



Mutually exclusive



Complement A^c

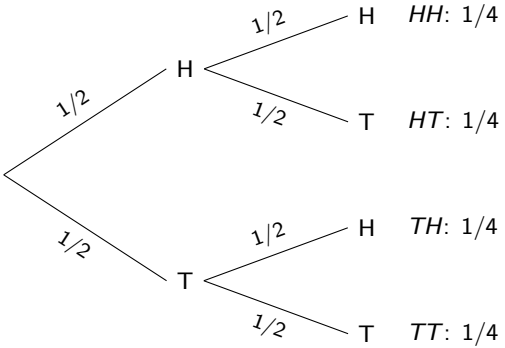


Throw two dice. Let A be "the first die is 1" and B be "the second die is 5."

- $A = \{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6)\}$ and $B = \{(1,5),(2,5),(3,5),(4,5),(5,5),(6,5)\}$.
- $A \cap B = \{(1,5)\}$, so $A \cup B$ has $6 + 6 - 1 = 11$ outcomes.

Roll one die. $A = \{1,3,5\}$ (odd) and $B = \{2,4,6\}$ (even) are mutually exclusive, and $A^c = B$.

Probability Tree: Two Coin Tosses



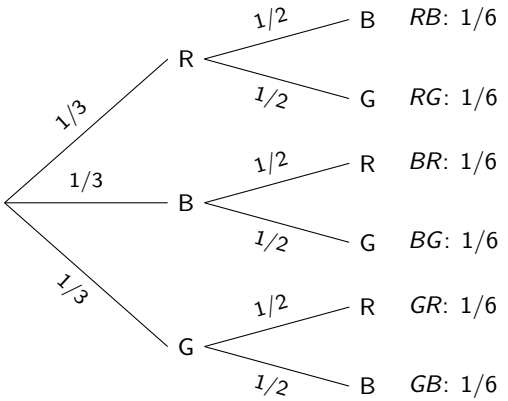
A probability tree represents the events of an experiment by lines. Toss a fair coin twice.

The probability of one H and one T: $P(HT) + P(TH) = 1/2$.

The probability of two T: $P(TT) = 1/4$.

What about the probability of at least one H?

Probability Tree: Selection Without Replacement



A bowl contains three M&Ms: one red (R), one blue (B), and one green (G). A child selects two at random without replacement.

The probability of no R:
$P(BG) + P(GB) = 1/3$.

What about the probability of at least one R?

The Axioms of Probability

- Probability distribution: A function that assigns a number, the probability, to each possible event.
- All valid probability distributions obey three axioms:
  1. Probabilities are never negative: $P(A) \geq 0$.
  2. One of the outcomes will definitely happen: $P(\Omega) = 1$.
  3. For any two disjoint events A and B: $P(A \cup B) = P(A) + P(B)$. $A \cup B$: A union B
- All other properties of probability can be derived from these three axioms.

Some Rules of Probability

Let A and B be two events. The three axioms imply:

- Probabilities cannot be higher than one: $0 \leq P(A) \leq 1$.
- The probability of nothing happening is zero: $P(\emptyset) = 0$.
- Complement rule: $P(A^c) = 1 - P(A)$.
- Identical events have the same probability, and if $A \subset B$ (belongs to), then $P(A) \leq P(B)$.
- Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
  If A and B are mutually exclusive, $P(A \cap B) = 0$ and $P(A \cup B) = P(A) + P(B)$.

Fair Roulette: An Example

Conditional = Joint / marginal

- Assume that the wheel is fair: each outcome has the same probability, p. This is an assumption, but a reasonable one because casinos are required by law to run fair wheels.
- Axiom 2 says $P(\Omega) = 1$. Since the outcomes are disjoint, axiom 3 says

$$P(\Omega) = P(\{0\}) + P(\{1\}) + \cdots + P(\{36\}) = 37p = 1.$$

  So $p = 1/37 \approx 0.027$.

- To find the probability of an event, add up the probabilities of its elementary events: $P(A) = |A| \times 1/37$.
- $P(Red) = 18/37 \approx 0.486$. What about $P(First12)$, where $First12 = \{1, 2, \ldots, 12\}$? $12/37 \approx 0.324$.

$|A|$ = number of outcomes lead to event A

Rules of Probability: Examples

- Roll a die. Let $A =$ "roll an even number" $= \{2, 4, 6\}$, so $P(A) = 3/6 = 1/2$. Then $P(A^c) = 1 - 1/2 = 1/2$.
- Roll a die. $P(\text{roll a } 10) = P(\emptyset) = 0$.
- An invoice is paid on time (A) with probability 0.65 and paid late (B) with probability 0.20. It cannot be both, so

$$P(A \cup B) = 0.65 + 0.20 = 0.85.$$

- Roll a die. Let $A = \{2, 4, 6\}$ and $B = \{3, 4, 5, 6\}$. Then $A \cap B = \{4, 6\}$ and

$$P(A \cup B) = 1/2 + 2/3 - 1/3 = 5/6.$$

Joint Probability

- We are often interested in how two events are related, e.g., a poker player after seeing the cards, or an economist predicting future economic conditions from current data.
- Joint probability: The probability that both A and B happen, $P(A \cap B)$.
- Roulette: What is the probability that the ball lands on a number that is both red and even?

$$P(Red \cap Even) = P(\{12, 14, 16, 18, 30, 32, 34, 36\}) = 8/37 \approx 0.216$$

- Joint probabilities are just probabilities, so they obey all the rules of probability.

Contingency Table: An Example

분할표

$P(\text{Man}) = \frac{227}{478}$ $P(\text{Ski}) = \frac{247}{478} = \frac{141}{478}, \frac{90}{478}$

Events can be placed in a contingency (two-way) table. Responses of 478 customers:

Gender	Skis	Camera	Bike	Total
Man	117	50	60	227
Woman	130	91	30	251
Total	247	141	90	478

각 범위 전체 부분

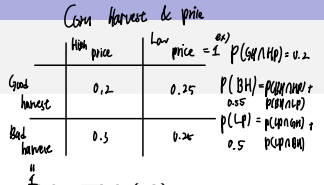
- Marginal probability (from the margins of the table): $P(\text{woman}) = 251/478 = 0.525$.
- Joint probability: $P(\text{woman} \cap \text{camera}) = 91/478 = 0.190$.
- Addition rule: $P(\text{woman} \cup \text{camera}) = (251 + 141 - 91)/478 = 0.63$.

MP written as Joint probability

$P(\text{woman}) = P(\text{woman} \cap \text{skis}) + P(\text{w} \cap \text{c}) + P(\text{w} \cap \text{B})$

$P(\text{Man} \cap \text{Ski}) = \frac{117}{478}, \frac{50}{478}, \frac{60}{478}$

Contingency Table: Cable Services



A cable company has 38 million cable passings. 10.9 million have only cable TV (A), 10.1 million have only cable internet (B), and 8.2 million have both.

	Internet, B	No internet, B^c	Total
Television, A	8.2	10.9	19.1
No television, A^c	10.1	8.8	18.9
Total	18.3	19.7	38

- Either service: $P(A \cup B) = (19.1 + 18.3 - 8.2)/38 \approx 0.77$.
- What about neither service? $P(A^c \cap B^c) = 8.8/38 = 0.23$.

Conditional Probability

- **Conditional probability:** The probability of A given that B has already occurred,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$

- It answers the question: if we already know that B is true, what are the chances that A is true?
- Examples: $P(\text{Recover from disease} \mid \text{Take the medicine})$, $P(\text{Hired} \mid \text{Pass the interview})$.
- Poker: At the start, every player has the same chance of winning. After you see your cards, your bet should be based on $P(\text{win} \mid \text{cards})$, not $P(\text{win})$.

Conditional Probability: Examples

	Remote work	In person	
w-f	0.25	0.45	0.7
w-p	0.15	0.15	0.3
	0.4	0.6	

$P(w\text{-}p \mid \text{in-person}) = \frac{0.15}{0.6}$

- Roulette:

$$P(Red \mid Even) = \frac{8/37}{18/37} \approx 0.444, \qquad P(b = 14 \mid Even) = \frac{1/37}{18/37} \approx 0.056$$

- Remote work: Let $A =$ "works remotely" and $B =$ "works full-time," with $P(A) = 0.40$, $P(B) = 0.70$, and $P(A \cap B) = 0.25$.

$$P(A \mid B) = \frac{0.25}{0.70} \approx 0.36$$

- Mutual funds: Let $A =$ "manager graduated from a top-20 MBA program" and $B =$ "fund outperforms the market," with $P(A \cap B) = 0.11$ and $P(A) = 0.40$.

$$P(B \mid A) = \frac{0.11}{0.40} = 0.275$$

Multiplication Rule

$P(A|B) = \frac{P(A \cap B)}{P(B)}$ / $P(A \cap B) = P(A|B) \cdot P(B)$ / $P(B \cap A) = P(B|A) \cdot P(A)$

- Rearranging the conditional probability gives the multiplication rule:

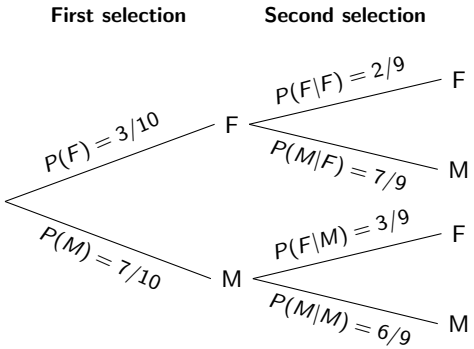
$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A).$$

- Some useful identities: (complement)
  - $P(A) = P(A \cap B) + P(A \cap B^c)$
  - $P(A \cup B) = 1 - P(A^c \cap B^c)$ and $P(A \cap B) = 1 - P(A^c \cup B^c)$
- In a company, 20% of employees work in IT (I), 30% hold a certification (C), and 50% of IT employees hold a certification. Then

$$P(I \cap C) = P(C \mid I)\, P(I) = 0.5 \times 0.2 = 0.1, \qquad P(I \mid C) = \frac{0.1}{0.3} = 0.33.$$

What about $P(I \cap C^c)$? $0.2 - 0.1 = 0.1$.

Probability Trees



- First branch: marginal probability. Second branch: conditional probability. Their product: joint probability.
- The branches from one node must add up to 1.
- A class has 7 male and 3 female students. Two are chosen without replacement. What is the probability that both are female?

$$P(F \cap F) = \frac{3}{10} \times \frac{2}{9} = \frac{6}{90}$$

Independent Events

- Independence ($A \perp B$): A and B are independent if

$$P(A \cap B) = P(A)\,P(B).$$

$P(A|B) = P(A)$ (ii) $P(A \cap B) = P(A)P(B)$

(i) $\frac{P(A \cap B)}{P(B)} = P(A)$

- If both events have nonzero probability, independence implies $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$. Knowing that one event is true tells you nothing about the other.
- Independence is reasonable when there is no physical or logical relationship between the events, usually because they are separated in time and space.
- Independence is not the same as mutually exclusive. Disjoint events have $P(A \cap B) = 0$.

Independence Across Roulette Games

Example Joint two Heads, four two coin toss

$P(H_1 \cap H_2) = P(H_1) \times P(H_2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

- Outcomes of two different roulette games can reasonably be assumed to be independent.
- You bring \$100 and bet everything on Red. If you win, you keep everything on Red for the next spin. After 3 games, you have \$800 if Red wins all 3 times, and \$0 otherwise.
- Because the games are independent,

$$P(w = \$800) = P(Red_1)\,P(Red_2)\,P(Red_3) = (18/37)^3 \approx 0.115.$$

- What about $P(w = \$0)$? $1 - 0.115 = 0.885$.

Checking Independence

- Events defined on the same outcome are typically related.
- Roulette: $P(Red) = 18/37 \approx 0.486$, but $P(Red \mid Even) = 8/18 \approx 0.444$. So Red and Even are not independent.
- Mutual funds: $P(B \mid A) = 0.275$, but $P(B) = 0.17$. So a top-20 MBA and outperforming the market are not independent.
- IT and certification: $P(I \mid C) = 0.33$, but $P(I) = 0.20$. Not independent.
- Independence is an assumption, and one that can easily be incorrect. Do not use results that only apply under independence when there is no reason to believe it holds.

Law of Total Probability

$$P(A) = \frac{P(A \cap B) \cdot P(B)}{P(B)} + \frac{P(A \cap B^c) P(B^c)}{P(B^c)}$$

$$= P(A \cap B) + P(A \cap B^c)$$

	H.P	L.P
GH	0.2	0.25
BH	0.3	0.25

$$P(GH \mid HP) = \frac{0.2}{0.5}$$

$$P(GH \mid HP^c) = \frac{0.25}{0.5}$$

- The law of total probability gives an unconditional probability from conditional probabilities:

$$P(A) = P(A \mid B)\, P(B) + P(A \mid B^c)\, P(B^c).$$

$$P(GH) = \frac{0.2}{0.5} \times 0.5 + \frac{0.25}{0.5} \times 0.5 = 0.45$$

- More generally, if $S_1, \ldots, S_k$ are mutually exclusive and exhaustive,

$$P(E) = P(E \mid S_1)P(S_1) + P(E \mid S_2)P(S_2) + \cdots + P(E \mid S_k)P(S_k).$$

- We create a set of scenarios, calculate the probability under each scenario, and add them up. This is useful for outcomes that occur in multiple stages.

Poker: An Example

- You are playing Texas hold'em, and one card is left to deal.
- If the last card is a heart (25% probability), you will have a flush and win with probability 90%. If not, you will win with probability 10%.
- Your overall chance of winning is

$$\begin{aligned} P(Win) &= P(Win \mid Hearts)\, P(Hearts) + P(Win \mid \text{not } Hearts)\, P(\text{not } Hearts) \\ &= 0.9 \times 0.25 + 0.1 \times 0.75 = 0.3. \end{aligned}$$

Bayes' Law

- Bayes' law restates one conditional probability in terms of the other:

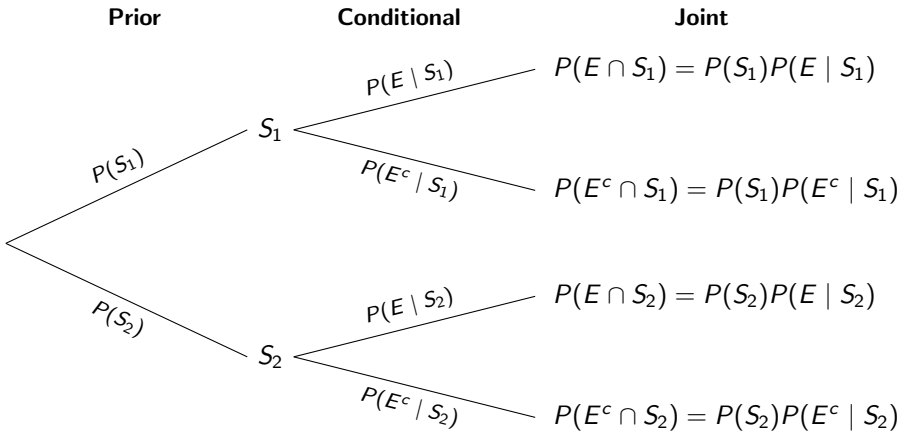
$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}.$$

- It is useful for evaluating evidence. Both Bayes' law and the law of total probability follow from the definition of conditional probability.
- More generally, with $S_1, \ldots, S_k$ mutually exclusive and exhaustive,

$$P(S_i \mid E) = \frac{P(E \mid S_i)\, P(S_i)}{\sum_{j=1}^{k} P(E \mid S_j)\, P(S_j)}.$$

$P(S_i)$ is the prior probability, $P(E \mid S_i)$ is the likelihood, and $P(S_i \mid E)$ is the posterior probability.

Bayes' Law: A Tree Diagram



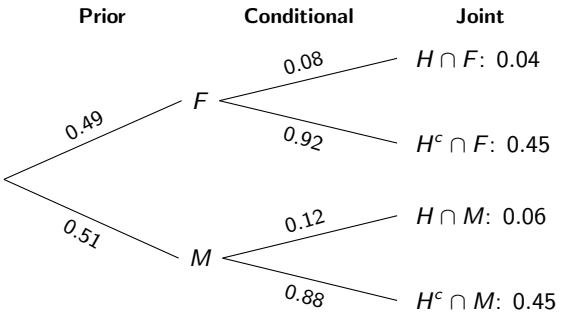
Medical Testing: An Example

- Let $T =$ "the patient tests positive" and $D =$ "the patient has the disease." No test is perfect.
- Sensitivity: $P(T \mid D) = p$. Specificity: $P(T^c \mid D^c) = q$. Prevalence: $P(D) = d$.
- A patient tests positive. What is $P(D \mid T)$? Bayes' law and the law of total probability give

$$P(D \mid T) = \frac{P(T \mid D)\,P(D)}{P(T)} = \frac{pd}{pd + (1-q)(1-d)}.$$

- Suppose $p = 1$ and $q = 0.99$. If the disease is common ($d = 0.10$), $P(D \mid T) \approx 0.917$. What about a rare disease ($d = 0.001$)? $P(D \mid T) \approx 0.091$.
- The same test means very different things depending on prevalence. A small false positive rate matters a lot when prevalence is low.

Heart Attack Risk: An Example



49% of the population are female. 8% of females and 12% of males are high risk for a heart attack (H).

A randomly selected person is high risk. What is the probability that this person is male?

$$P(M \mid H) = \frac{P(H \cap M)}{P(H \cap M) + P(H \cap F)}$$

$$= \frac{0.06}{0.06 + 0.04} = 0.6$$

Quality Control: An Example

- A company has three production lines. Line S_1 produces 70% of output with a 4% defect rate, S_2 produces 20% with a 2% defect rate, and S_3 produces 10% with a 96% defect rate.
- A randomly selected product is defective (H). What is the probability that it came from S_1?
- Joint probabilities: $P(H \cap S_1) = 0.028$, $P(H \cap S_2) = 0.004$, $P(H \cap S_3) = 0.096$.
- Law of total probability: $P(H) = 0.028 + 0.004 + 0.096 = 0.128$.
- Bayes' law: $P(S_1 \mid H) = 0.028/0.128 \approx 0.219$, down from the prior of 0.70.

Summary

- Probability rules: $0 \leq P(A) \leq 1$, $P(A^c) = 1 - P(A)$, and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Conditional probability: $P(A \mid B) = P(A \cap B)/P(B)$, and multiplication rule: $P(A \cap B) = P(A \mid B)\, P(B)$.
- Independence: $P(A \cap B) = P(A)\, P(B)$, equivalently $P(A \mid B) = P(A)$. Independence is not the same as mutually exclusive.
- Law of total probability: $P(E) = \sum_i P(E \mid S_i)\, P(S_i)$.
- Bayes' law: $P(S_i \mid E) = P(E \mid S_i)\, P(S_i)/P(E)$.

Practice Question 1: Dish Detergent

$P(A) = 0.65$ $P(B) = 0.48$ $P(A \cap B) = 0.25$

	A	A^c	
B	0.25	0.23	0.48
B^c	0.4	0.12	0.52
	0.65	0.35	

A company launched two new dish detergents. Among sampled customers, 65% liked detergent A, 48% liked detergent B, and 25% liked both.

(a) State the three probabilities in terms of the events A and B.

(b) Construct a 2×2 contingency table with rows A, A^c and columns B, B^c.

(c) What is the probability that a customer likes only detergent A?

(d) What is the probability that a customer likes at least one detergent?

(e) What is the probability that a customer likes neither detergent?

(f) What is the probability that a customer likes only one detergent?

(g) Are A and B mutually exclusive? NO. $P(A \cap B) = 0.25$

Practice Question 1: Answers

(a) $P(A) = 0.65$, $P(B) = 0.48$, $P(A \cap B) = 0.25$.

(b)

	B	B^c	Total
A	0.25	0.40	0.65
A^c	0.23	0.12	0.35
Total	0.48	0.52	1

(c) $P(A \cap B^c) = P(A) - P(A \cap B) = 0.40$.

(d) $P(A \cup B) = 0.65 + 0.48 - 0.25 = 0.88$.

(e) $P(A^c \cap B^c) = 1 - P(A \cup B) = 0.12$.

(f) $P(A \cap B^c) + P(A^c \cap B) = 0.40 + 0.23 = 0.63$.

(g) No, because $P(A \cap B) = 0.25 \neq 0$.

Practice Question 2: Credit Score and Loan Default

In a survey, 60% of people had a good credit score. Of those with a good credit score, 8% defaulted on a loan. Of those without a good credit score, 25% defaulted.

Let C = "has a good credit score" and D = "defaulted on a loan."

(a) Express the three probabilities in terms of C and D.

(b) What is the probability that a randomly selected person defaulted on a loan?

(c) If a person defaulted on a loan, what is the probability that the person did not have a good credit score?

Practice Question 2: Answers

(a) $P(C) = 0.60$, $P(D \mid C) = 0.08$, $P(D \mid C^c) = 0.25$.

(b) Law of total probability:

$$\begin{aligned} P(D) &= P(D \mid C)P(C) + P(D \mid C^c)P(C^c) \\ &= 0.08 \times 0.60 + 0.25 \times 0.40 = 0.048 + 0.10 = 0.148 \end{aligned}$$

(c) Bayes' law:

$$P(C^c \mid D) = \frac{P(C^c \cap D)}{P(D)} = \frac{0.10}{0.10 + 0.048} \approx 0.68$$

Practice Question 3: The Pass Bet

The main bet in craps is Pass. On the first ("come out") roll c, Pass wins if c is 7 or 11 and loses if c is 2, 3, or 12. Otherwise c becomes the "point," and the shooter keeps rolling: Pass wins if the point comes up again before a 7, and loses otherwise.

c	2	3	4	5	6	7	8	9	10	11	12
$P(c)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$
$P(PassWins \mid c)$	0	0	$\frac{3}{9}$	$\frac{4}{10}$	$\frac{5}{11}$	1	$\frac{5}{11}$	$\frac{4}{10}$	$\frac{3}{9}$	1	0

Use the law of total probability to calculate $P(PassWins)$.

Practice Question 3: Answer

$$
\begin{aligned}
P(PassWins) &= P(PassWins \mid c=2)P(c=2) + \cdots + P(PassWins \mid c=12)P(c=12) \\
&= \left(0 \times \tfrac{1}{36}\right) + \left(0 \times \tfrac{2}{36}\right) + \left(\tfrac{3}{9} \times \tfrac{3}{36}\right) + \left(\tfrac{4}{10} \times \tfrac{4}{36}\right) + \left(\tfrac{5}{11} \times \tfrac{5}{36}\right) + \left(1 \times \tfrac{6}{36}\right) \\
&\quad + \left(\tfrac{5}{11} \times \tfrac{5}{36}\right) + \left(\tfrac{4}{10} \times \tfrac{4}{36}\right) + \left(\tfrac{3}{9} \times \tfrac{3}{36}\right) + \left(1 \times \tfrac{2}{36}\right) + \left(0 \times \tfrac{1}{36}\right) \\
&= \frac{244}{495} \approx 0.4929
\end{aligned}
$$

- This is less than 50%, so the house still has an advantage.
- But it is higher than the probability of winning a bet on Red in roulette, $18/37 \approx 0.486$.

Practice Question 4: Returning Planes

During World War II, the statistician Abraham Wald analyzed bullet damage on returning planes. A plane can be hit on the engine ($EngineHit$) or on the body ($BodyHit$), and it either returns safely ($Return$) or crashes. Suppose that:

- $P(EngineHit) = P(BodyHit) = 0.6$
- Many of the returning planes show body damage: $P(BodyHit \mid Return) = 0.7$
- Few of the returning planes show engine damage: $P(EngineHit \mid Return) = 0.1$
- Most of the planes return: $P(Return) = 0.8$

Calculate $P(Return \mid BodyHit)$ and $P(Return \mid EngineHit)$.

Practice Question 4: Answer

Bayes' law:

$$P(Return \mid BodyHit) = \frac{P(BodyHit \mid Return)\,P(Return)}{P(BodyHit)} = \frac{0.7 \times 0.8}{0.6} \approx 0.93$$

$$P(Return \mid EngineHit) = \frac{P(EngineHit \mid Return)\,P(Return)}{P(EngineHit)} = \frac{0.1 \times 0.8}{0.6} \approx 0.13$$

- A plane hit on the body usually returns, but a plane hit on the engine usually does not.